

Layperson's Summary

It is 2026 and we exist in an interconnected world. The connectivity of our human experience is however not a new observation, in the early 20th century Hungarian poet and writer Frigyes Karinthy produced a short story centered on this idea of an interconnected world. In his 1929 short story, Chains (Láncszemek), he suggested the global population at the time of 1.5 Billion people were at most five acquaintances away from one another. This idea was popularised in John Guare's 1990 play "Six Degrees of Separation".

In 2026, however, compared with 1929 or even 1990, we exist in an interconnected world that leaves an abundance of measurable footprints. These footprints exist in the digital, in the form of data, and allow us to study this connectivity more rigorously than ever before. In 2011, using this data, an experiment was performed, finding that the average degree of separation on Twitter between two users was 3.84 [1]. In other words, researchers found the shortest chain of mutual connections typically involved fewer than four accounts. So, if written just two decades later, Guare's play would need renaming.

Amazingly, the mathematics underlying this connectivity is strikingly similar across a diverse range of systems, from molecular biology to computer science and beyond [2, 3]. When we describe such systems in terms of the connections between their constituent elements, regardless of whether they are molecules or Twitter accounts, we refer to them as networks. These networks can have very different and often complex structures. However, almost magically, many exhibit properties that can be captured and compared using the exact same mathematical equations. Such equations can offer insight into questions ranging from how biological interactions shape genetic evolution to how many acquaintances I might need before I can become friends with an Arsenal player. In this work, one such equation is created to answer the more broadly applicable question:

Given the characteristics of my network, how likely is a member within it to attract the number of connections it does?

The answer presented follows from the fact that, for a wide variety of networks, growth when new members are added is accurately described by just two processes: already well-connected members tend to attract more connections [4, 5, 6], and members with similar characteristics are more likely to connect [7, 8]. Ultimately, this work is the creation of a new equation to describe how networks grow based solely on these two processes.

To provide a more concrete example, this work focuses on networks of scientific publications (though our results would remain unchanged if we instead considered other networks, from social systems [8] to food webs [9]). In this context, one way to reframe the question above is:

Given the characteristics of an academic paper (for example, the gender of its author and its field of research), how likely is it that the paper will be referenced, say, one hundred times by other academic papers?

Given the right data (often the limiting factor), the equation created in this work can answer exactly this question. But to what degree of accuracy?

Finally, this work creates a new statistical test to assess to what extent the predictions of this equation can be relied upon. Using this test, it was found that (aside from the fact that it is not the most efficient way to make friends with Arsenal players), for the majority of simulated networks, this equation is exceedingly accurate.

References

- [1] Reza Bakhshandeh et al. ‘Degrees of Separation in Social Networks’. In: *Proceedings of the International Symposium on Combinatorial Search*. Vol. 2. 1. 2021, pp. 18–23.
- [2] R. Gallagher and T. Appenzeller. ‘Beyond Reductionism’. In: *Science* 284 (1999), p. 79.
- [3] R. F. Service. ‘Exploring the Systems of Life’. In: *Science* 284 (1999), p. 80.
- [4] A.-L. Barabási and R. Albert. ‘Emergence of scaling in random networks’. In: *Science* 286.5439 (1999), pp. 509–512.
- [5] Derek J. de Solla Price. ‘A general theory of bibliometric and other cumulative advantage processes’. In: *Journal of the American Society for Information Science* 27.5 (1976), pp. 292–306.
- [6] Robert K. Merton. ‘The Matthew Effect in Science’. In: *Science* 159.3810 (1968), pp. 56–63.
- [7] Miller McPherson, Lynn Smith-Lovin and James M. Cook. ‘Birds of a Feather: Homophily in Social Networks’. In: *Annual Review of Sociology* 27 (2001), pp. 415–444.
- [8] Fariba Karimi et al. ‘Homophily influences ranking of minorities in social networks’. In: *Scientific Reports* 8.1 (2018), p. 11077.
- [9] Ferenc Jordán, Zsófia Benedek and János Podani. ‘Quantifying positional importance in food webs: A comparison of centrality indices’. In: *Ecological Modelling* 205.1 (2007), pp. 270–275.